How Is the Brand Effect of Vehicles on Sales in America

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1. Introduction

1.1. Background

The automotive industry represents a highly competitive and capital-intensive sector, where purchasing decisions involve significant financial commitments and long-term considerations. In such markets, brand characteristics often play a pivotal role alongside objective product attributes such as price, performance, and size. Previous studies suggest that brands may function as signals of quality, helping consumers mitigate information asymmetry and uncertainty in durable goods markets. In addition, brand loyalty has been recognized as a critical factor influencing repeat purchases and sustaining market share. While considerable research has examined the determinants of vehicle sales, much of the existing literature has focused primarily on price, technical features, and macroeconomic conditions. The independent role of brand identity—separated from product specifications—has received comparatively less systematic investigation. This study addresses this gap by using a dataset of conventional fuel vehicles from multiple manufacturers, allowing for the analysis of brand effects on sales while controlling for key product and performance variables.

1.2. Literature Review

Numerous studies have explored the influence of brand-related factors on automobile sales. Key determinants such as brand value, brand loyalty, and brand image have been identified as significant contributors to consumer purchasing decisions and

overall sales performance.

1. Brand Value and Vehicle Sales

Nadanyiova et al. (2019) conducted a study to define the theoretical basis of brand value and its impact on sales in the automotive industry. Their analysis revealed a medium-strong direct linear dependence between brand value and sales, indicating that higher brand value positively affects sales volume.

A case study by Desklib analyzed the impact of brand value on sales across five automotive companies. The findings underscored the importance of brand value in driving sales performance, suggesting that companies with higher brand value tend to achieve better sales outcomes.

2. Brand Loyalty and Market Share

J.D. Power(2024) reported that Toyota and Honda achieved the highest loyalty rates among mass-market brands, with 62.5% and 58.8% respectively. In the premium segment, Porsche led with a 57.5% loyalty rate. These figures highlight the critical role of brand loyalty in maintaining and expanding market share.

3. Brand Image and Consumer Purchase Behavior

Antara and Rastini (2022) investigated the influence of brand image, sales promotion, and service quality on customer satisfaction in the car rental service industry. Their study concluded that brand image has a positive and significant impact on customer satisfaction, emphasizing the necessity for companies to cultivate a strong brand image to enhance customer experiences.

Qianling’s(2023) study examined the impact mechanism of consumers' initial visits to

automobile 4S stores on test drive intentions. The research found that brand image significantly influences consumers' willingness to test drive, indicating that a positive brand image can effectively encourage potential customers to engage with the product.

1.3. Research Significance

Understanding how automotive brands affect vehicle sales is important both theoretically and practically. Theoretically, brands signal quality and influence consumer utility beyond price and specifications, helping explain how brand equity drives market advantages. Practically, knowing how brand attributes shape sales can guide firms in resource allocation, marketing, and competition strategies. While existing studies link brand value to sales, many do not fully isolate brand effects from product features, making it difficult to assess the independent influence of branding. This study addresses these gaps and offers new insights for both academic research and industry practice.

2. Data

2.1. Data source:

Kaggle (https://www.kaggle.com/competitions/nlp-getting-started/data)

The dataset contains 157 observations, with each observation representing a specific car model. It includes 16 variables covering brand identity, sales outcomes, technical specifications, and other relevant characteristics.

Table 1 summarizes the variables included in the dataset along with their descriptions:

Table 1 Description of the Variables

Variable Name	Description
Manufacturer(Brand)	Name of the car manufacturer (e.g., Ford, Toyota)
Model	Name of the specific car model
Sales_in_thousands	Vehicle sales, measured in thousands of units
Vehicle_type	Type of vehicle (e.g., SUV, Sedan)
Price_in_thousands	Vehicle price, measured in thousands of US dollars
Engine_size	Engine displacement, measured in liters
Horsepower	Engine power output, measured in horsepower
Wheelbase	Distance between the front and rear axles, measured in inches
Width	Vehicle width, measured in inches
Length	Vehicle length, measured in inches
Curb_weight	Vehicle curb weight, measured in pounds
Fuel_capacity	Fuel tank capacity, measured in gallons
Fuel_efficiency	Fuel efficiency, measured in miles per gallon (mpg)
Latest_Launch	Latest launch date of the vehicle (string format)
Power_perf_factor	Power performance index
resale_missing	Indicator for missing resale value data (0 = no missing, 1 = missing)

2.2. Dataset Advantage

The dataset presents several advantages for empirical analysis. First, it contains rich multidimensional information, including vehicle prices, engine specifications, fuel efficiency, and physical dimensions, allowing for careful control of product characteristics and mitigating omitted variable bias. Second, it covers a broad range of automobile brands, enabling cross-brand comparisons and facilitating the analysis of brand effects on sales outcomes. Third, the continuous measurement of vehicle sales (in thousands of units) permits precise modeling using linear regression techniques rather than coarse categorical comparisons. Fourth, the availability of both performance indicators and pricing information allows for clear separation between the effects of tangible product attributes and intangible brand value. Fifth, the dataset

structure—featuring standardized variables and minimal missing values—ensures analytical clarity and reduces data preparation time. Together, these features make the dataset particularly well-suited for identifying and quantifying the role of brand characteristics in shaping consumer purchasing behavior.

2.3. Descriptive Summary of the Data

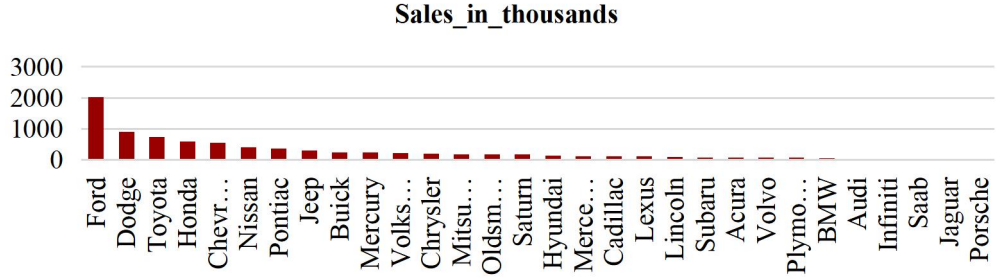
We began our analysis by examining the raw dataset, which includes 157 vehicle models from various manufacturers. The primary variable of interest is Sales_in_thousands, representing the annual sales volume for each model. Initial summary statistics revealed substantial variability in both vehicle specifications and sales outcomes.

Table 2. Descriptive Statistic Summary

	Sales_in_thousands	Price_in_thousands	Engine_size	Horsepower
AVG	52.99	27.39	3.06	185.95
MAX	540.56	85.5	8	450
MIN	0.11	9.235	1	55

Ford stands out as the top performer, with an total sales volume of approximately 2,000,000, followed by Honda and Jeep. In contrast, luxury brands such as Porsche and Saab recorded significantly lower average sales. The distribution of Sales_in_thousands is notably right-skewed, indicating that a small number of high-selling models contribute disproportionately to overall sales.

Figure 1. Sales of each brand



2.4. Data Cleaning

Before conducting any modeling, we addressed missing values to ensure data quality. For variables with fewer than 5% missing observations—such as Fuel_efficiency, Curb_weight, and Price_in_thousands—we applied median imputation. This method allowed us to maintain dataset size while minimizing the impact of outliers.

The variable X__year_resale_value had a missing rate of around 23%, making it unsuitable for imputation. Instead, we created a binary indicator (resale_missing) to capture the presence of missing values, and then removed the original column from further analysis. This approach allowed us to retain potential insights from missing data without introducing noise. Additionally, categorical variables like Manufacturer were converted to factor types for use in modeling.

Table 3. Variables Missing Rate

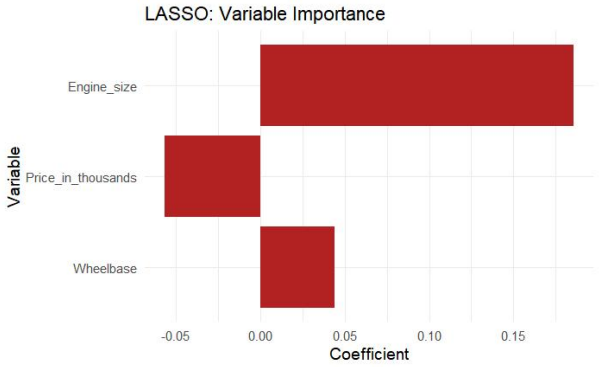
Variable	Missing_Percent(%)
X__year_resale_value	22.93
Fuel_efficiency	1.91
Price_in_thousands	1.27
...	...

3. Empirical Analysis

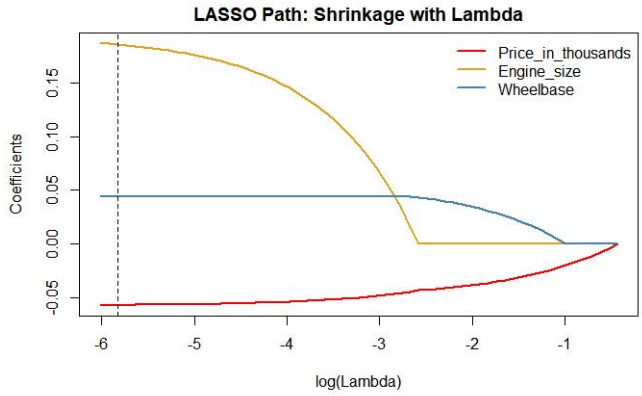
3.1. LASSO Regression

To identify the most important predictors of car sales, we implemented LASSO (Least Absolute Shrinkage and Selection Operator) regression. We used 10-fold cross-validation to determine the optimal penalty parameter, ensuring reliable and generalizable variable selection.

The final LASSO model retained three variables with non-zero coefficients:



These results are consistent with economic expectations. Vehicles with larger engines tend to attract higher demand, while more expensive models typically sell in smaller volumes. The LASSO coefficient path plot showed how other variables were gradually reduced to zero as the penalty increased, confirming LASSO's role in eliminating irrelevant or redundant predictors.



These selected variables were used in subsequent regression and classification models to examine brand-specific effects and categorize vehicles into different sales performance tiers.

3.2. Model Methods

When we go back to review the original data in chart 1, this data summary shows that the Ford had an average wheelbase of 112.7, 24.21 price in thousands and an average engine size of 3.33 turbo. Ford’s features are not the most outdstanding, cars that are made by other companies are in some perspectives better than Ford such as Dodge has

large wheelbase, Honda has lower price. It seems that Ford's cars are not enough to be the best seller. Base on these, here is a question that what makes consumers to buy Ford in such a big volume. The answer is the brand effect. In details, Ford is an American brand and will occupy the domestic market more and earlier, plus the much more investment in advertising and the government's protection of manufacturers of their own country, Ford has created a unbeatable brand and continually affected people in consuming in vehicles.

	Stratified by Manufacturer					
	level	Acura	Audi			
n		4	3			
Wheelbase (mean (SD))		107.70 (5.50)	108.10 (5.23)			
Price_in_thousands (mean (SD))		28.67 (9.37)	39.98 (19.71)			
Engine_size (mean (SD))		2.93 (0.76)	2.93 (1.21)			
	Stratified by Manufacturer					
	BMW	Buick	Cadillac			
n	3	4	5			
Wheelbase (mean (SD))	108.67 (2.37)	111.00 (2.40)	112.08 (4.42)			
Price_in_thousands (mean (SD))	33.10 (5.96)	26.78 (4.22)	40.25 (5.90)			
Engine_size (mean (SD))	2.70 (0.17)	3.62 (0.35)	4.50 (0.96)			
	Stratified by Manufacturer					
	Chevrolet	Chrysler				
n	9	7				
Wheelbase (mean (SD))	103.93 (5.92)	109.10 (3.87)				
Price_in_thousands (mean (SD))	20.02 (10.56)	23.34 (4.50)				
Engine_size (mean (SD))	3.06 (1.34)	2.81 (0.56)				
	Stratified by Manufacturer					
	Dodge	Ford				
n	11	11				
Wheelbase (mean (SD))	114.67 (12.78)	112.70 (11.32)				
Price_in_thousands (mean (SD))	24.21 (15.47)	21.05 (8.04)				
Engine_size (mean (SD))	3.71 (1.82)	3.33 (1.03)				
	Stratified by Manufacturer					
	Honda	Hyundai	Infiniti			
n	5	3	1			
Wheelbase (mean (SD))	107.56 (6.14)	100.93 (5.12)	108.30 (NA)			
Price_in_thousands (mean (SD))	20.28 (6.16)	12.17 (2.67)	29.46 (NA)			
Engine_size (mean (SD))	2.52 (0.80)	1.97 (0.45)	3.00 (NA)			

Chart 1: Summary of the features' data

To quantitatively go over the brand effect, this paper use 'Base Line' test which is a OLS regression model with dummy variables. Here is the basic formula of this model.

Sales = β_0 + β_1·Honda +⋯+ β_k·Toyota + γ·Price + δ·Engine_Size+θ ·Wheelbase +ϵ

In this model, Audi, Toyota, Honda and other names of the car brand (expect of Ford)

are the Brand Dummy Variables. Dummy Variable equals to 1 if the name is correctly matched the brand while if the name is not matched, dummy variables remain 0. The reason that there is no Ford Dummy Variable is we set β_0 as the Baseline Sales of Ford and β_1 to β_k are the sales difference between other manufacturers and Ford. Applying this model aims to test if there are still sales gap between Ford and other brand regardless of all the features' effect. In other words, if all the cars have the same level of price, wheelbase and engine size, which brand can be the best seller.

Chart 2 shows the result of this regression model. As we can see, all the coefficient β_k is negative, which means every car brand has a significant sales gap with Ford. The coefficient of Jeep is the biggest and Honda occupies the second place. It indicates that except of Ford, Jeep is the manufacturer with the most powerful brand effect and then is Honda. However, when we go back to see the original data, Jeep is not the second popular car brand in America, instead, the second place belongs to Honda, that is because when consumers compare Jeep and Honda, features effect is authorized, the good features, price, engine size and wheelbase drive consumers to buy Honda's car more than Jeep.As for Ford, it has unbeatable brand effect to guarantee its sales.

```
Residuals:
     Min       1Q   Median       3Q      Max
-132.634  -21.216   -2.058   15.101  271.962

Coefficients:
                    Estimate Std. Error t value Pr(>|t|)
(Intercept)         -196.131     77.555  -2.529 0.012696 *
ManufacturerAcura   -137.109     30.780  -4.455 1.85e-05 ***
ManufacturerAudi    -130.758     36.958  -3.538 0.000568 ***
ManufacturerBMW     -139.822     35.586  -3.929 0.000141 ***
ManufacturerBuick   -109.486     29.702  -3.686 0.000339 ***
ManufacturerCadillac -133.170    28.673  -4.645 8.56e-06 ***
ManufacturerChevrolet  -91.983    23.491  -3.916 0.000148 ***
ManufacturerChrysler -139.989    25.120  -5.573 1.49e-07 ***
ManufacturerDodge   -103.717     21.664  -4.787 4.72e-06 ***
ManufacturerHonda    -48.984     28.006  -1.749 0.082761 .
ManufacturerHyundai -108.590     34.116  -3.183 0.001843 **
ManufacturerInfiniti -134.223    53.545  -2.507 0.013480 *
ManufacturerJaguar  -148.599     56.174  -2.645 0.009218 **
ManufacturerJeep     -40.304     34.369  -1.173 0.243162
ManufacturerLexus   -122.773     29.842  -4.114 7.03e-05 ***
ManufacturerLincoln -136.138     34.180  -3.983 0.000115 ***
ManufacturerMercedes-B -108.004  32.328  -3.341 0.001104 **
ManufacturerMercury -134.970     25.753  -5.241 6.65e-07 ***
ManufacturerMitsubishi -122.551  25.495  -4.807 4.35e-06 ***
ManufacturerNissan  -105.641     24.953  -4.234 4.44e-05 ***
ManufacturerOldsmobile -140.579  25.849  -5.438 2.75e-07 ***
ManufacturerPlymouth -157.225    30.612  -5.136 1.06e-06 ***
ManufacturerPontiac -106.578     25.835  -4.125 6.73e-05 ***
ManufacturerPorsche  -58.356     46.469  -1.256 0.211550
ManufacturerSaab    -134.770     41.783  -3.226 0.001608 **
ManufacturerSaturn  -129.249     28.406  -4.550 1.26e-05 ***
ManufacturerSubaru  -103.749     39.897  -2.600 0.010442 *
ManufacturerToyota   -72.221     23.869  -3.026 0.003016 **
ManufacturerVolkswagen -108.850  27.824  -3.912 0.000150 ***
ManufacturerVolvo   -130.027     29.758  -4.370 2.60e-05 ***
Wheelbase              3.658      0.728   5.025 1.72e-06 ***
Price_in_thousands    -1.241      0.751  -1.653 0.100945
Engine_size           -1.845      8.804  -0.210 0.834386
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 50.6 on 124 degrees of freedom
Multiple R-squared:  0.5603,    Adjusted R-squared:  0.4468
F-statistic: 4.937 on 32 and 124 DF,  p-value: 5.488e-11
```

Chart 2:Results of Model

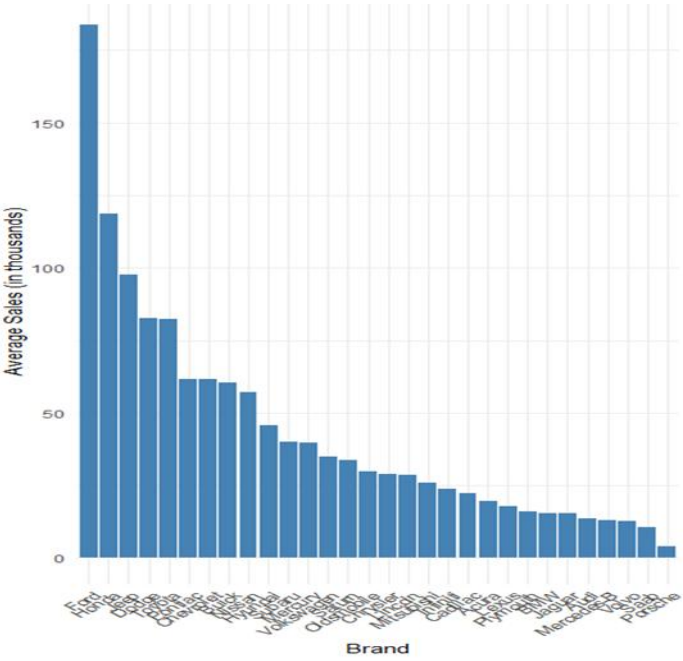


Figure 4:Average Sales by Brand

3.3. Logistic Regression

This section assesses the impact of incorporating brand information into the

classification of U.S. vehicle sales performance (High/ Medium/ Low). We compare two feature sets: one using only technical attributes (Price in thousands, Engine size, Wheelbase), and another adding manufacturer-level dummy variables. Both models are estimated using multinomial logistic regression and random forest, evaluated with 5-fold cross-validation based on accuracy and Cohen's Kappa.

$$\text{Sales Level}_i = \beta_0 + \beta_1 \times \text{Price}_i + \beta_2 \times \text{Engine Size}_i + \beta_3 \times \text{Wheelbase}_i + \sum_k \gamma_k \times \text{Brand Dummy}_{ik} + \epsilon_i$$

Logistic regression outperforms random forest in both metrics, with an average cross-validation accuracy of 64.27% and a Kappa of 0.4653 (vs. 55.44% and 0.3322, respectively), and is selected as the final model.

```
Accuracy
                Min.   1st Qu.    Median      Mean   3rd Qu.      Max. NA's
Logistic     0.5909091 0.6086957 0.6363636 0.6426877 0.6818182 0.6956522    0
RandomForest 0.5000000 0.5000000 0.5217391 0.5537549 0.5652174 0.6818182    0

Kappa
                Min.   1st Qu.    Median      Mean   3rd Qu.      Max. NA's
Logistic     0.3869969 0.4152542 0.4584615 0.4653092 0.5232198 0.5426136    0
RandomForest 0.2461059 0.2484472 0.2853107 0.3275389 0.3390805 0.5187500    0
```

Given these results, the logistic regression model is selected as the final specification. On the holdout test set, this model delivers an overall accuracy of 53.33% and a Kappa of 0.3. Disaggregated performance metrics reveal that the model correctly classifies 8 out of 15 High-sales vehicles (recall = 53.33%), with a balanced accuracy of 70% for this class.

```
Confusion Matrix and Statistics

          Reference
Prediction Low Medium High
    Low      9      6    1
    Medium   4      7    6
    High     2      2    8
```

```
Statistics by Class:

                     Class: Low Class: Medium Class: High
Sensitivity              0.6000        0.4667       0.5333
Specificity              0.7667        0.6667       0.8667
Pos Pred Value           0.5625        0.4118       0.6667
Neg Pred Value           0.7931        0.7143       0.7879
Prevalence               0.3333        0.3333       0.3333
Detection Rate           0.2000        0.1556       0.1778
Detection Prevalence     0.3556        0.3778       0.2667
Balanced Accuracy        0.6833        0.5667       0.7000
```

To further evaluate performance, we simplify the task to a binary classification of High vs. Non-High and compute the ROC curve. The resulting AUC of 0.8244 implies that in a random High/Non-High pair, the model has an 82.44% chance of correctly ranking the High-sales vehicle higher.

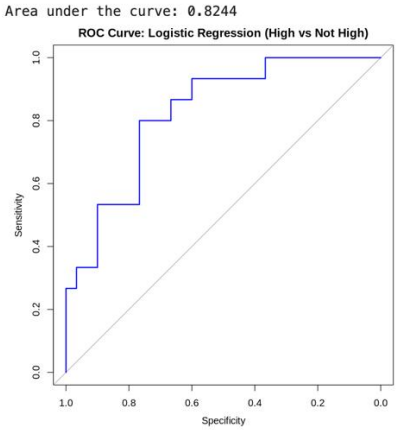


Figure 5. ROC Curve

3.4. Comparison

Model Version	Feature Set	Cross-Validation Mean Accuracy	Test Set Accuracy	Test Set AUC
Technical Features Only	Price, Engine Size, Wheelbase	55.44%	53.33%	0.798
Technical + Brand Features	Price, Engine Size, Wheelbase + Brand Dummies	64.27%	53.33%	**0.8244**

Table 4. Comparison between Two Model Version

By comparison, the model trained exclusively on technical features achieves a lower cross-validation accuracy (53.33%) and test-set AUC (0.798). This consistent

improvement in both performance metrics affirms the predictive salience of brand identity. Overall, the inclusion of brand dummies materially enhances the model's classification performance and facilitates more accurate identification of top-selling vehicles—offering insights that go beyond what can be inferred from technical specifications alone.

4. Conclusion

4.1. Confirming the Brand Premium Effect

Our analysis confirms the existence of a significant brand premium effect in the U.S. automotive market. Even after accounting for fundamental technical attributes (such as price, engine size, and wheelbase), brand identity emerges as a strong positive predictor of vehicle sales. In both the linear (OLS) and logistic regression models, many brand dummy variables showed significant coefficients, indicating that vehicles from certain brands tend to achieve higher sales than comparable models from lesser-known brands. This finding aligns with the product-market perspective of brand equity, where a branded offering delivers enhanced sales performance relative to an equivalent unbranded product. In practical terms, a strong brand reputation provides a measurable uplift in sales beyond what technical specifications alone would predict, validating the notion of a brand-driven sales premium in the automotive sector.

4.2. Brand Identity and Technical Features in Combination

While brand identity plays a critical role, our results also underscore the importance of technical features in driving sales. Core attributes like price, engine capacity, and design characteristics remain fundamental determinants of a car model's market performance. However, the interplay between these tangible features and the intangible brand factor is where the greatest insight emerges. Our models indicate that technical merits and brand reputation *jointly* influence consumer purchase decisions. For instance, two vehicles with similar specifications can have markedly different sales outcomes if one carries a stronger brand badge. Thus, brand identity complements technical features, suggesting that consumers evaluate both the product's attributes and the prestige or trust associated with its brand. This synergy means that neither aspect alone is sufficient; instead, integrating brand factors with vehicle specifications provides the most comprehensive understanding of sales drivers.

4.3. Limitations

A notable limitation of our model is its difficulty in distinguishing between medium and low sales categories. While the inclusion of brand and technical features enabled clear identification of top-selling models, the predictive separation between mid-tier and low-tier sales performers was less reliable. In the logistic classification context, many models in the middle range of sales were often misclassified or grouped with low-selling models, indicating that the features used did not capture sufficiently fine-grained differences. This could be due to overlapping characteristics between

medium-selling and low-selling vehicles or noise in the sales data. In practical terms, the model's resolution is higher at the extremes (successes vs. failures) but blurs in the middle. This limitation suggests caution in interpreting the model's predictions for moderate sales levels. It also indicates that additional predictive features or alternative modeling approaches might be needed to improve granularity – for example, incorporating variables that better discriminate mid-range popularity, or using a multi-class classification technique tailored to handle subtle distinctions among sales categories.

Rreference

Nadanyiova, M., Gajanova, L., Majerova, J., & Nadanyiova, M. (2019). The Brand Value and its Impact on Sales in Automotive Industry.

Desklib. (n.d.). Case Study: Brand Value Impact on Automotive Sales. Desklib Desklib

J.D. Power. (2024). 2024 U.S. Automotive Brand Loyalty Study.

Antara, I. M. R. S., & Rastini, N. M. S. (2022). The Influence of Brand Image, Sales Promotion and Quality of Service on Customer Satisfaction Car Rental Service.

Qianling Jiang, Liyuan Deng,& Chun Yang. (2023). The Impact Mechanism of Consumer's Initial Visit to an Automobile 4S Store on Test Drive Intention: Product Aesthetics, Space Image, Service Quality, and Brand Image.